

New Insight into the Confinement Effect of Microporous Carbon in Li/Se Battery Chemistry: A Cathode with Enhanced Conductivity

Xiwen Wang, Yuqing Tan, Zhixiao Liu, Yuqin Fan, Mingnan Li, Hussein A. Younus, Junfei Duan, Huiqiu Deng, and Shiguo Zhang*

Embedding the fragmented selenium into the micropores of carbon host has been regarded as an effective strategy to change the Li–Se chemistry by a solid–solid mechanism, thereby enabling an excellent cycling stability in Li–Se batteries using carbonate electrolyte. However, the effect of spatial confinement by micropores in the electrochemical behavior of carbon/selenium materials remains ambiguous. A comparative study of using both microporous (MiC) and mesoporous carbons (MeC) with narrow pore size distribution as selenium hosts is herein reported. Systematic investigations reveal that the high Se utilization rate and better electrode kinetics of MiC/Se cathode than MeC/Se cathode may originate from both its improved Li⁺ and electronic conductivities. The small pore size (<1.35 nm) of the carbon matrices not only facilitates the formation of a compact and robust solid-electrolyte interface (SEI) with low interfacial resistance on cathode, but also alters the insulating nature of Li₂Se due to the emergence of itinerant electrons. By comparing the electrochemical behavior of MiC/Se cathode and the matching relationship between the diameter of pores and the dimension of solvent molecules in carbonate, ether, and solvate ionic liquid electrolyte, the key role of SEI film in the operation of C/Se cathode by quasi-solid-solid mechanism is also highlighted.

1. Introduction

With the development of portable devices and electric vehicles, the lithium-ion batteries that consist of graphite anode and lithium transition metal-oxide cathode cannot fulfill the

increasing requirement of energy density. Among these alternatives to the conventional lithium-ion batteries, lithium–sulfur (Li–S) batteries are of particular interest due to their high theoretical gravimetric energy density (2500 Wh kg^{−1}) and natural abundance of sulfur.^[1] Selenium (Se), as an element that also belongs to the chalcogen group in periodic table, has been recently proposed as a promising cathode material.^[2–5] In spite of the lower theoretical gravimetric capacity of Se (675 mAh g^{−1}), the volumetric capacity of Se (3253 mAh cm^{−3}) is comparable to that of S (3467 mAh cm^{−3}). Moreover, Se possesses an electronic conductivity that is ≈20 orders of magnitude higher than S, which enables a higher utilization rate, better kinetics and higher loading of active materials in electrode. These advantages make Se a promising candidate of cathode materials for high energy lithium battery.

The Li–Se electrochemistry highly depends on the organic electrolyte. In

ether-based electrolytes, the Se cathode operated via solid–liquid–solid mechanism suffers from the dissolution of reaction intermediates–lithium polyselenide, resulting in the continuous leakage of active materials and shuttle reaction between cathode and anode.^[4–6] Using the carbonate-based electrolytes that seldom dissolve lithium polyselenide intermediates is a potential strategy to address this problem occurred in Li–Se batteries.^[7] However, because of the irreversible reactions between the nucleophilic long-chain polysulfide anions (Se^{2−} or Se^{2−}) and carbonate molecules, the bulk Se cathode even prepared with nanostructured Se active materials is incompatible with these electrolytes.^[5,8,9] Recently, it was found that the Se confined in CMK-3 matrix can perform well with carbonate electrolytes.^[2] By forming the space-confined Se chains in pores of carbon hosts, the Se molecules can be directly converted to Li₂Se without the lithium polyselenide intermediates.^[10] Moreover, the terminal Se molecules in helical Se_x are preferentially attacked by Li⁺, enabling a higher electrochemical activity than the ring structure.^[11] Inspired by this work, various carbon hosts have also been demonstrated for Li–Se batteries. Compared to the mesoporous carbon/Se (MeC/Se)^[12–14] and hierarchical porous carbon/Se cathodes,^[15–17] the MiC/Se cathodes such as metal-organic framework-derived MiC,^[18–20] graphitic MiC

Dr. X. Wang, Y. Tan, Dr. Z. Liu, Y. Fan, M. Li, Dr. H. A. Younus, Prof. S. Zhang
College of Materials Science and Engineering
State Key Laboratory of Advanced Design and Manufacturing for Vehicle Body
Hunan University
Tianma Rd. 27, Changsha, Hunan 410082, China
E-mail: zhangsg@hnu.edu.cn

Dr. J. Duan
School of Materials Science and Engineering
Changsha University of Science and Technology
Changsha, Hunan 410004, China

Dr. H. Deng
School of Physics and Electronics
Hunan University
Changsha, Hunan 410082, China

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/smll.202000266>.

DOI: 10.1002/smll.202000266

sheets^[21] and MiC sphere^[22,23] shows obviously higher specific capacity, better cycling stability, along with better rate capability. However, the chain-like Se molecules can be formed in pores which are usually not larger than 5 nm.^[2,5] Its formation could not be the main reason responsible for the better performance of MiC/Se composites. Therefore, a deeper insight into the reaction pathways of the MiC confined Se in carbonate electrolyte is therefore highly required. Besides, it is well-known that the solvated ionic liquid shows some special characteristics, such as high ionic conductivity, wide electrochemical window, well compatibility with lithium polysulfide and low solubility of intermediate lithium polysulfide.^[24–26] Nevertheless, to our knowledge, no work about the solvated ionic liquid electrolyte in Li–Se batteries has been reported.

In this work, two porous carbon hosts with different pore sizes (average 1.02 and 4.35 nm) were designed to encapsulate Se, and the relationship between the pore structure of carbon hosts and the electrode kinetics of the corresponding Se cathodes was systematically investigated. It is revealed that the high Se utilization rate and better electrode kinetics of MiC/Se cathode arise from its both improved ionic and electronic

conductivities. By combining the experimental analysis and theoretical calculations, the underlying mechanism of the enhanced conductivity is further explored. In addition, the electrochemical behavior of MiC/Se cathode in various electrolytes including carbonate, ether and solvate ionic liquid is analyzed. We believe that the new findings about the effect of nanoconfinement on Li/Se chemistry may not only broaden our understanding of the unprecedented electrochemical phenomena, but also provide guidance to the rational design of metal–Se batteries.

2. Results and Discussion

As illustrated in **Figure 1a**, microporous carbon (MiC) was synthesized by a simple approach using NaY zeolite as a hard template. The rhombic shape of NaY zeolite with 200–300 nm size is well preserved in MiC during the carbonization process, as shown in **Figure 1b,c**. The high-resolution transmission electron microscopy (HRTEM) images clearly confirm the existence of micropores and their uniform distribution in MiC (**Figure S1a–c**, Supporting Information). Owing to its intrinsic properties like ordered porous

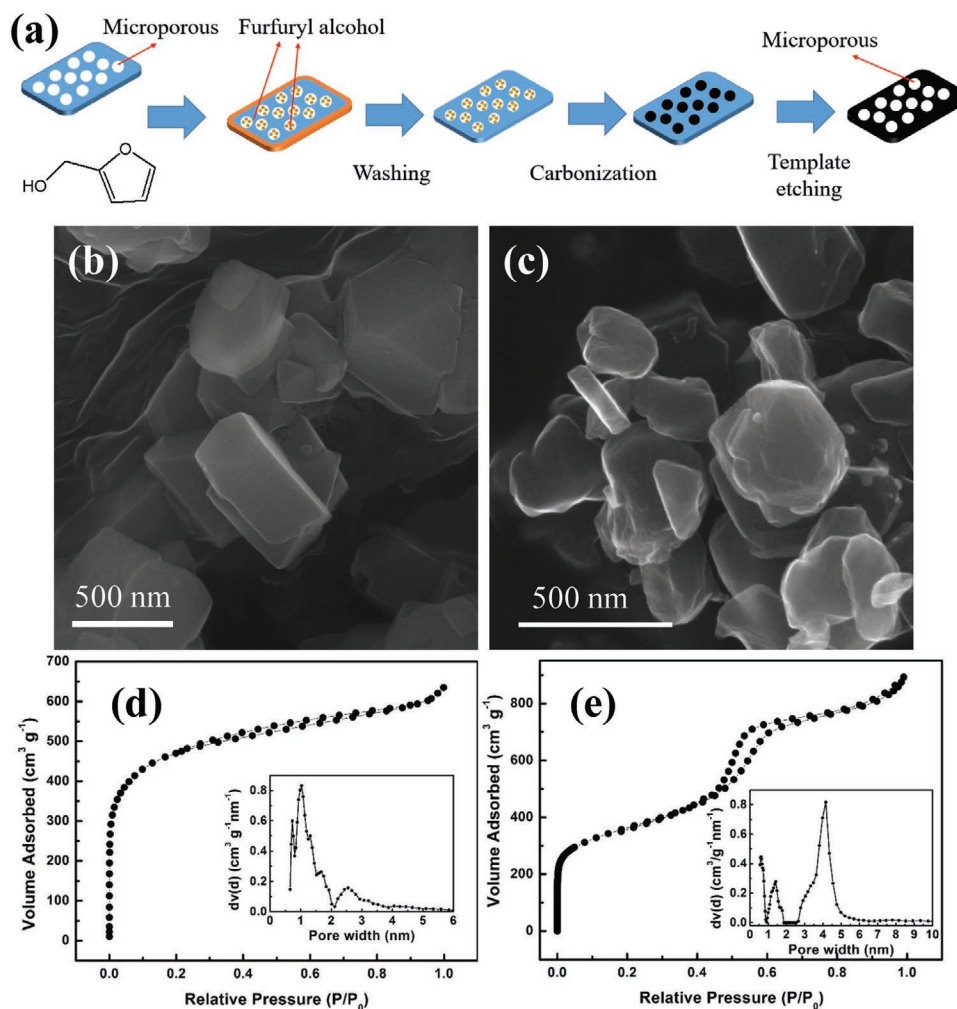


Figure 1. Structure of microporous and mesoporous carbon Se hosts. a) Schematic illustration of the synthesis process of MiC materials. SEM images of b) NaY zeolite template and c) MiC materials. N₂ adsorption/desorption isotherms of d) MiC materials and e) MeC materials. Inset of (d) and (e) are the corresponding pore size distribution plots.

Table 1. Comparison of specific surface area and pore volume of the carbon and carbon/selenium materials.

Materials	Specific surface area [$\text{m}^2 \text{g}^{-1}$]	Pore volume [$\text{cm}^3 \text{g}^{-1}$]
MiC (NaY templated carbon)	1757	0.85
MeC (CMK-3)	1265	1.38
N-MeC	1115	1.24
MiC/Se composites	22	0.05

channels and high electronic conductivity, CMK-3, which has been widely employed as a mesoporous host to accommodate active material for Li-S and Li-Se battery in previous reports,^[2,27,28] was also selected as the MeC host in this work to demonstrate the effect of pore size on the electrochemical behavior and mechanism. To further analyze the various pore features of MiC and MeC, N_2 adsorption-desorption isotherms were tested (Figure 1d,e). As can be seen from Figure 1d,e, MiC shows a type I isotherm, while that of MeC is type IV (Figure 1d,e), indicating the microporous and mesoporous structure for MiC and MeC, respectively. Correspondingly, the pore size distributions of MiC and MeC are mainly concentrated at 1.02 and 4.15 nm, which match well with initial design. As expected, MiC possesses a larger specific surface area ($1757 \text{ m}^2 \text{g}^{-1}$) but smaller pore volume ($0.85 \text{ cm}^3 \text{g}^{-1}$) than MeC ($1265 \text{ m}^2 \text{g}^{-1}$ and $1.38 \text{ cm}^3 \text{g}^{-1}$), as summarized in Table 1.

Elemental Se was then uniformly infiltrated into the pores of MiC and MeC via a facile melt-diffusion strategy, without

changing the morphology of the carbon host (Figure S1d–f, Supporting Information). The Se content of MiC/Se and MeC/Se determined by thermo-gravimetry analysis (TGA) are 44.2 and 40.5 wt%, respectively (Figure S2, Supporting Information). It should be noted that the Se content in this work was intentionally controlled to prevent the agglomeration of Se on the carbon surface and ensure enough space available to buffer the volume change of Se/ Li_2Se reaction in pores, based on the calculated Se content under full- Li_2Se filling condition (for details see supporting information). An interesting phenomenon drawn from the TGA result (Figure S2, Supporting Information) is that the weight loss temperature of Se at the temperature range of 400–600 °C significantly increased when changing the sample from bulk Se to MeC/Se and MiC/Se. The much higher temperature required to evaporate confined Se for C/Se is due to a large driving force to overcome the strong capillary force of pores in carbon hosts. After infiltrating Se into the nanoporous carbon materials, the characteristic X-ray diffraction (XRD) patterns of the trigonal phase of Se disappear (Figure 2a), indicating the amorphous nature of Se in both MiC/Se and MeC/Se composites. Raman spectra of Se, MiC/Se and MeC/Se were collected to further investigate the Se structure in MiC and MeC composites (Figure 2b). Even though the incorporation of Se has no influence on the relative intensity of the D and G peaks and thus the graphitization degree of carbon materials, a noticeable blue shift of the characteristic band of Se is observed from 236 for bulk Se to 258 cm^{-1} for both MiC/Se and MeC/Se (Figure 2c). This result suggests not only the

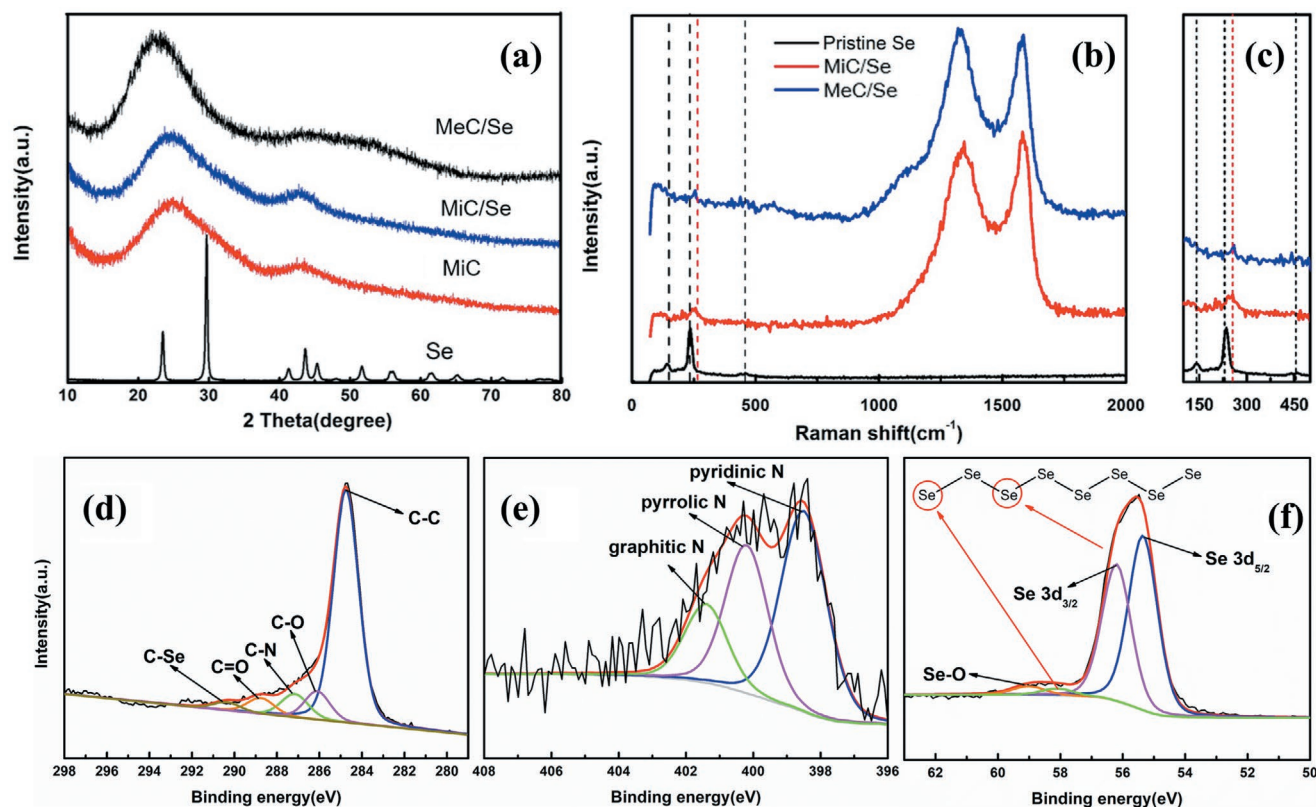


Figure 2. Properties of C/Se composites. a) XRD patterns, b) Raman spectra, and c) amplified Raman spectra ($100\text{--}500 \text{ cm}^{-1}$) of pristine Se, MiC/Se, and MeC/Se composites. d) C 1s, e) N 1s, and f) Se 3d XPS spectra of MiC/Se composites.

transformation of cyclo- Se_8 in the trigonal crystal to chain-like Se_x after infiltration into the nanopores,^[2,11,29] but also the similar molecular structure of Se in MiC and MeC host. According to previous reports, Se atoms in chain-like Se_x molecules have a stronger interaction with the carbon atoms of host than that in Se_8 rings.^[2,5] This concept was proven here by X-ray photo-electron spectroscopy (XPS) results (Figure 2d–f). The peaks located at 59.0 eV in Se 3d spectrum of MiC/Se were ascribed to the Se–O chemical bond.^[23,30,31] Apart from the typical peaks of Se 3d_{3/2} (55.3 eV) and Se 3d_{5/2} (56.2 eV), a weak peak at 58.1 eV, which can be attributed to the terminal Se atoms of chain-like Se_x molecules, as indication for the generation of Se chain in MiC.^[23] The N-doping in MiC is confirmed by the N 1s peak, which can be divided into the pyridine-N, pyrrole-N, and quaternary-N (Figure 2e).

All electrochemical tests are performed using the coin-cells with carbonate-based electrolytes. The cyclic voltammograms (CV) of MiC/Se cathode (Figure 3a) display a reduction peak at 1.70 V and an oxidation peak at 2.05 V. In consistent with

the CV results, the galvanostatic discharge–charge profiles of MiC/Se cathode (Figure 3b) show only an inclined plateau in lithiation or delithiation process, and similar phenomenon is also observed in the voltage profiles of MeC/Se cathode (Figure S3, Supporting Information). The single-plateau electrochemical behavior is related to the direct phase transformation from Se to Li_2Se in carbonate-based electrolyte, without forming polyselenides.^[10,11] The MiC/Se cathode demonstrates an initial capacity of 971 mAh g^{-1} with excellent cycling stability and rate capability (Figure 3c). The initial capacity beyond the theoretical value of 675 mAh g^{-1} can be ascribed to the high utilization of short order chain-like Se_x molecules and the irreversible formation of solid-electrolyte interface (SEI) on the surface of electrode in the initial cycle.^[4,20,32–34] After 500 cycles, the capacity maintains at 400 mAh g^{-1} at 0.5 C, with capacity decay rate of 0.082% per cycle (from 2nd to 500th cycle) and Coulombic efficiency of nearly 100%. At a high rate of 5 C, the MiC/Se cathode can still deliver a capacity of 363 mAh g^{-1} and it can recover to 644.6 mAh g^{-1} when the current rate returns to

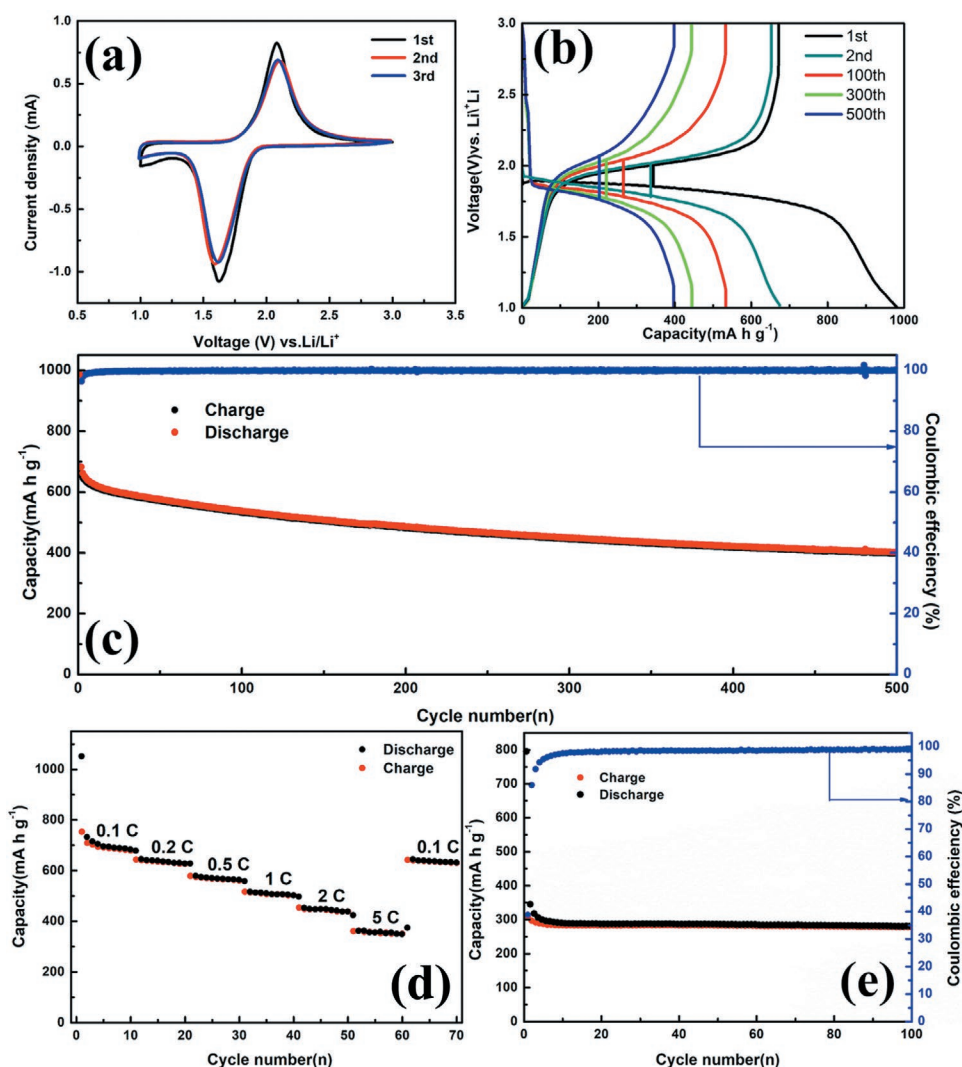


Figure 3. Electrochemical performance of MiC/Se cathode. a) CV curves at a scan rate of 0.1 mV s^{-1} . b) Voltage profiles of different cycles at 0.5 C. c) Cycling performance and Coulombic efficiency. d) Rate performance. e) Cycling performance of MeC/Se cathode at 0.5 C.

0.1 C (Figure 3d). In contrast, the MeC/Se cathode possesses the specific capacity lower than 300 mAh g^{-1} , though it maintains capacity decay of 0.064% per cycle (Figure 3e). In addition, the potential gap between charge and discharge curves of MiC/Se cathode (0.15 V for 2nd cycle) is obviously smaller than that of MeC/Se cathode (0.33 V for 2nd cycle), implying the considerable improvement of kinetics in MiC/Se cathode. We further increase the Se content of MiC/Se composite to 577 wt% that almost reaches full-Li₂Se filling condition and prepared the MiC/Se electrode with a higher areal Se loading of 2.2 mg cm^{-2} (Figure S4, Supporting Information). The MiC/Se cathode with high Se content shows an initial capacity of 945 mAh g^{-1} and retains a capacity of 348 mAh g^{-1} after 500 cycles, corresponding to a capacity decay rate of 0.094% per cycle (from 2nd to 500th cycle). Thus, the MiC carbon host in this work also holds a great promise for the practical feasible Li–Se batteries.

Undoubtedly, the above results confirmed the much better electrochemical performance of MiC/Se than that of MeC/Se. Especially for the solid–solid reaction pathway in carbonate electrolytes or all-solid state electrolyte, the impact of both Li⁺ and electronic conductivity are more profound.^[7,35–37] This may be also the reason that the MiC/Se cathodes exhibits lower polarization and better rate performance than MeC/Se cathode. To confirm this point, electrochemical impedance spectra of MiC/Se and MeC/Se cathodes before and after cycling were tested (Figure 4a,b). All the Nyquist impedance plots are composed of a depressed semicircle and a sloping line. According to the previous reports, the equivalent circuit of MiC/Se and MeC/Se cathode in carbonate electrolyte can be expressed as

the inset of Figure 4a.^[4,14,20,38] In the equivalent circuit, the R_s represents the resistance of electrolyte, R_{int} is the interface resistance including the charge transfer resistance and SEI resistance, W_o is the Warburg impedance due to the diffusion of the Li-ion within the pores of carbon hosts. The values of R_s and R_{int} for C/Se cathodes before and after cycling are listed in Table 2. After 20 cycles, the interface resistance of MiC/Se cathode (20.1Ω) is clearly smaller than that of MeC/Se cathode (44.9Ω), implying fast Li⁺ transfer through SEI. While the R_s of MeC cathode increased obviously after cycling (Figure S5, Supporting Information). Because the side reaction between the polyselenide anions and carbonate molecules consumes the solvent, the viscosity of electrolyte will increase. Compared to cell with MiC/Se cathode, the irreversible side reaction in cell with MeC/Se cathode is more severe, which resulted in a higher R_s for the cycled MeC cathode. In addition, the changed slope of linear region in MiC/Se cathode indicated a possible variation of pore tortuosity in carbon hosts, which have affected the Li⁺ diffusion within micropores. Furthermore, as indicated by the powder electrical resistivity tests under various pressures, the resistivity of MeC is close to that of MiC, and so are their corresponding C/Se composites (Figure S6, Supporting Information). It seems that the intrinsic electronic conductivity of C/Se cathode before cycling is almost unaffected by the pore size of carbon hosts. The Li⁺ diffusion coefficients (D_{Li^+}) of the cells with the two cathodes were then evaluated. By a series of CV measurements under various scanning rates from 0.05 to 0.5 mV s^{-1} , the D_{Li^+} can be calculated based on the Randles-Sevcik equation (Figure 4c; Figure S7, Supporting Information), which is also widely used in Li–S batteries^[39–41]

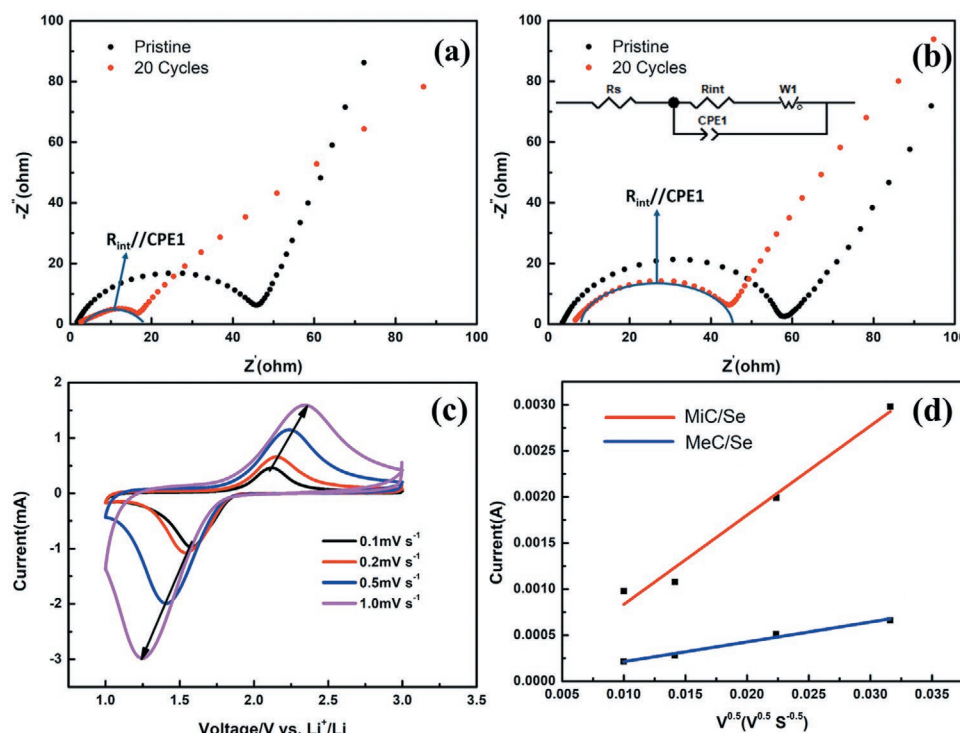


Figure 4. Conductivities of MiC/Se and MeC/Se cathodes. Nyquist plots of a) MiC/Se cathode and b) MeC/Se cathode before cycling and after 20 cycles at 0.5 C. c) CV curves of MiC/Se cathode with different scan rates: 0.1, 0.2, 0.5, and 1.0 mV s^{-1} . d) Plots of peak current of anodic reaction versus square root of the scan rate for the MiC/Se and MeC/Se cathodes.

Table 2. Fitted resistances data from the equivalent circuit of MiC/Se and MeC/Se cathode in carbonate electrolyte.

Cathode	Stage	R_s	R_{int}
MiC/Se	Before cycling	1.6	46.6
	After cycling	2.2	20.1
MeC/Se	Before cycling	3.2	54.5
	After cycling	6.0	44.9

$$I_p = (2.69 \times 10^5) n^{1.5} a D_{Li^+}^{0.5} C v^{0.5} \Delta C_o \quad (1)$$

where I_p is the peak current; Cv is the scan rate of CV measurements; n is the number of electrons per reaction species; a is the active electrode area; ΔC_o is the Li concentration change in electrolyte. The n , a , and ΔC_o are constant in this work. The D_{Li^+} calculated based on the cathodic peaks were estimated to be 5.88×10^{-10} and $9.69 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ for the cell with MiC/Se and MeC/Se cathode, respectively (Figure 4d). Obviously, the higher lithium ion diffusion coefficients of MiC/Se cathode can be attributed to its enhanced Li^+ conductivity, leading to its better electrochemical kinetics.

During the preparation of MiC hosts, both the impregnation of furfuryl alcohol into NaY and chemical vapor deposition (CVD) of acetonitrile were found to be essential for obtaining high microporosity and ordering of MiC.^[42] However, the CVD process may inevitably introduce the N-doping in the MiC materials. To further excludes the possible influence of N-doping on electrochemical performance of C/Se composite and thus ensures a better comparison of the effect of pore structure, CMK-3 was intentionally doped with N heteroatoms via a simple ionic liquid carbonization strategy (for details see supporting information). The N elements are found to be uniformly distributed in N-MeC, as shown in Figure S8a–c in the Supporting Information. It seems that the surface area and pore properties of CMK-3 mesoporous host are almost unaffected by the post carbonization of ionic liquid (Figure S8d, Supporting Information). By carefully controlling the relative content of ionic liquid precursor, the doped N content in N-MeC (1.37 at%) is controlled to be close to that in the MiC (1.35 at%), and the type and distribution of N configurations in two materials are also similar (Figure S8f, Supporting Information). Because of the limited doped N content in N-MeC, the resistivity of N-MeC is close to that of MeC. The N-MeC/Se cathode exhibits the similar electrochemical properties to MeC/Se cathode, in terms of reversible capacities, cycling stability and charge/discharge potential gaps (Figure S9, Supporting Information). Therefore, we can conclude that the determining factor in two C/Se composites is pore properties rather than N-doping.

As concluded from previous literatures concerning quasi-solid-state reaction of Se cathode, the side reactions and formation of SEI on Se cathode is responsible for the irreversible capacity loss between the initial and second discharge.^[2,20,33] Similar phenomena can also be observed for both MiC/Se and MeC/Se cathode, and the irreversible capacity loss of MeC/Se cathode (450 mAh g^{-1}) is much larger than that of MiC/Se cathode (300 mAh g^{-1}) (Figure 3b; Figure S6, Supporting Information), implying a more severe electrolyte decomposition and a thicker SEI layer on the surface of MeC/Se cathode.

The surface morphology of cycled MiC/Se cathodes appear to be smooth and flat (Figure 5a,b). In contrast, scanning electron microscopy (SEM) images of MeC/Se cathode after 100 cycles at 0.5 C exhibit rough surface morphology with some apparent cracks (Figure 5a,b), which is associated with the incompact SEI film. HRTEM images (Figure 5e,f) further reveals a relatively thin (3–4 nm) and homogenous SEI layer on the C/Se particles in MiC/Se cathode after two cycles, while a significantly thicker (over 20 nm) and rough SEI layer is observed on the the MeC/Se particles. As a result, lower interface resistance and higher Li^+ ionic conductivity can be attained in MiC/Se cathode.

To further gain insights into the electrochemical process of different C/Se cathodes, we have studied the XPS spectra of C/Se cathode after one cycle. In the F 1s spectra (Figure 6a,b), the main peak at 685.0 eV comes from the LiF, while the additional peak at 687.2 eV can be assigned to the decomposition products of $LiPF_6$, such as Li_xPF_y and $Li_xPO_yF_z$.^[20,43,44] A more intense LiF peak in F 1s spectrum of MiC/Se cathode corresponds to SEI film with higher proportion of LiF, which is believed to be more dense, stable and protective. The compact SEI layer not only allows fast Li ions transport, but also provide long-term chemo-mechanical stability to preserve electrical contacting, namely the accessibility of active selenium species throughout the entire cathode. As for C 1s spectra of cycled MiC/Se cathode (Figure 6c,d), the relative intensity of the peak at 290.2 eV that corresponds to C–Se bond is obviously increased, indicating that the interaction between Se atoms in chain-like Se_x molecules and C atoms in sp^2 framework of MiC substrate become stronger after one cycle.^[23,45,46] Consistent with the C 1s spectra, an additional peak at 57.8 eV can be seen in Se 3d spectra due to the aforementioned C–Se bond (Figure 6e). However, the C–Se interaction in cycled MeC/Se cathode is greatly weaker than that of cycled MiC/Se cathode. In Se 3d spectra, two key difference between cycled MiC/Se and MeC/Se cathodes can be observed, i.e., the presence of Li 1s peaks at 55.7 eV and extra peaks at lower binding energy of 52–54 eV (Figure 6f). These peaks can be attributed to the Li_2Se and reduced selenium species, respectively.^[3,4] After a discharge–charge cycle at 3.0 V, the residue of Li_2Se and reduced selenium suggest the poor reversibility of the Se/ Li_2Se conversion reaction in initial discharge–charge process, which leads to low capacity and large over-potential in MeC/Se cathode. In addition, the Se–O species related to the peak at high binding energy of 59.3 eV may be ascribed to the side reaction between Se anions and the carbonyl group of carbonate solvent.^[4,23] However, the unfavorable side reaction is suppressed in MiC/Se cathode as evidenced by the absence of Se–O species in the corresponding Se 3d spectra.

A previous theoretical study on Na–Se battery demonstrated that nanoconfinement effect can enhance the electronic conductivity of $NaSe_x$.^[47] In this work, the electronic structure of nanoconfined Li_2Se is also studied. The nanoconfinement environment is modeled by inserting a $Li_{16}Se_8$ -cluster into SWCNTs with various diameters. Similar to the $Na_{16}Se_8$ -cluster confined in SWCNTs, it is found that the $Li_{16}Se_8$ -clusters incorporate with each other and form a one-dimensional Li_2Se nanowire in the interior space of SWCNTs. Although Li_2Se is more electronically conductive than Li_2S , the crystalline Li_2Se is still the insulator for transferring electrons. However, it is found that $Li_{16}Se_8$

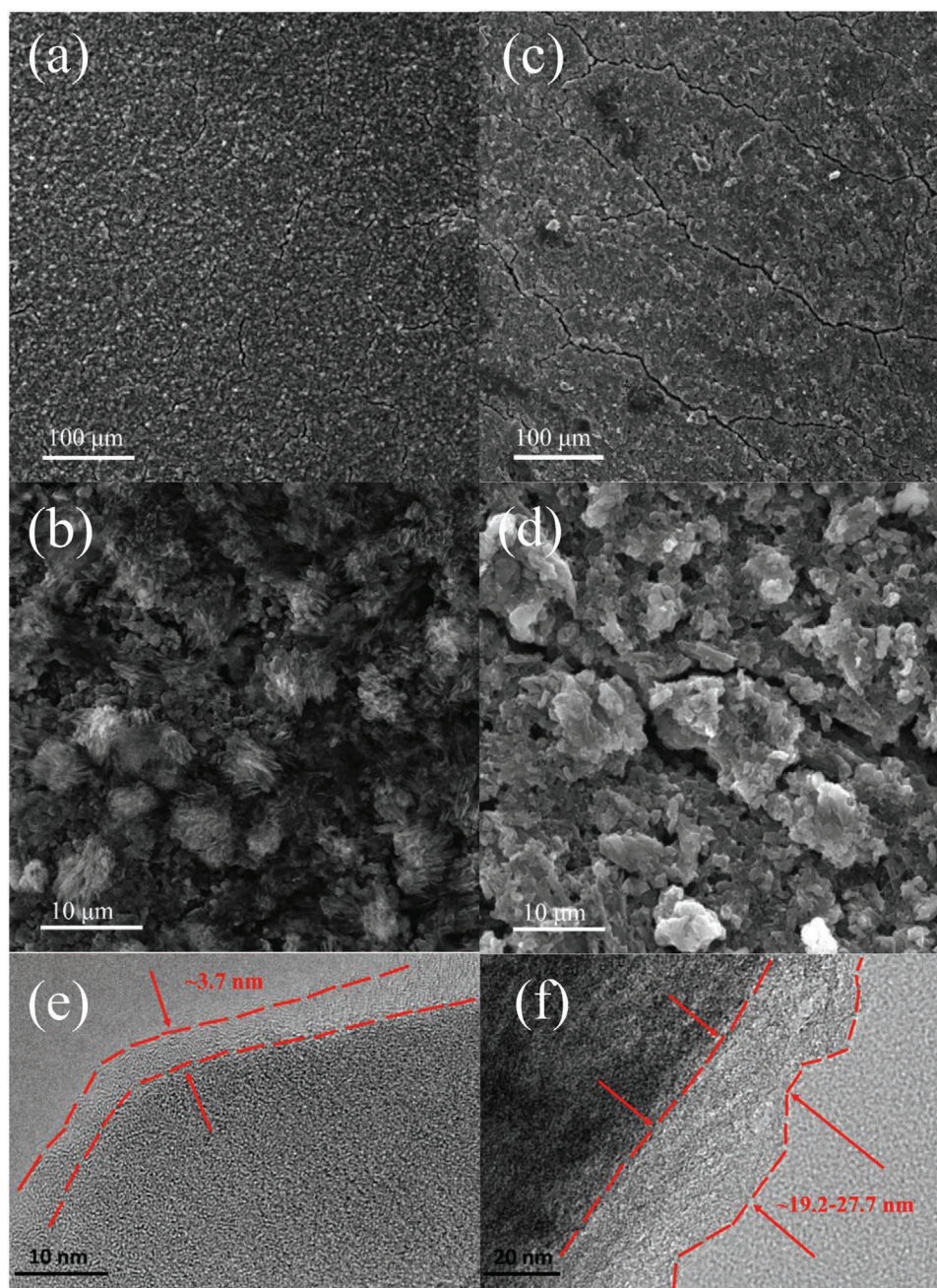


Figure 5. Morphological observations of SEI layers. SEM images of a,b) MiC/Se electrode and c,d) MeC/Se electrode after 100 cycles at 0.5 C. HRTEM images of e) MiC/Se and f) MeC/Se composites after the fifth cycle in carbonate-based electrolyte.

in the (10, 10) SWCNT (13.56 Å diameter) exhibits the metallic property (Figure 7a). By analyzing the total density of state (TDOS), the Fermi energy is above the conduction band minimum (CBM), as shown in Figure 7d. In contrast, larger (12, 12) SWCNT and (14, 14) SWCNT (Figure 7b,c) cannot induce itinerant electrons. As shown in Figure 6e,f, the band gap between the CBM and the valence band maximum (VBM) is too wide to transfer electrons. Therefore, the nanoconfinement effect of micropores, which alters the electronically insulating nature of Li_2Se , is one of the main reasons for the high specific capacity of MiC/Se cathode.

Along with the carbonate electrolyte, the electrochemical behaviors of MiC/Se cathode in including ether and solvate ionic liquid are also investigated. The cycled MiC/Se cathodes were first soaked in various electrolytes for 12 h (Figure 8a). The wine-red color of ether-based electrolyte solution indicates that Se molecules can dissolve in ether solvent and migrate out of MiC/Se cathode, despite the physical confinement of micropores in MiC and the chemical interaction between lithium polyselenides and N-containing functional groups. Whereas the carbonate-based electrolyte and solvate ionic-liquid electrolyte ($[\text{Li}(\text{G4})][\text{TFSI}]$) remain colorless. Accordingly, the

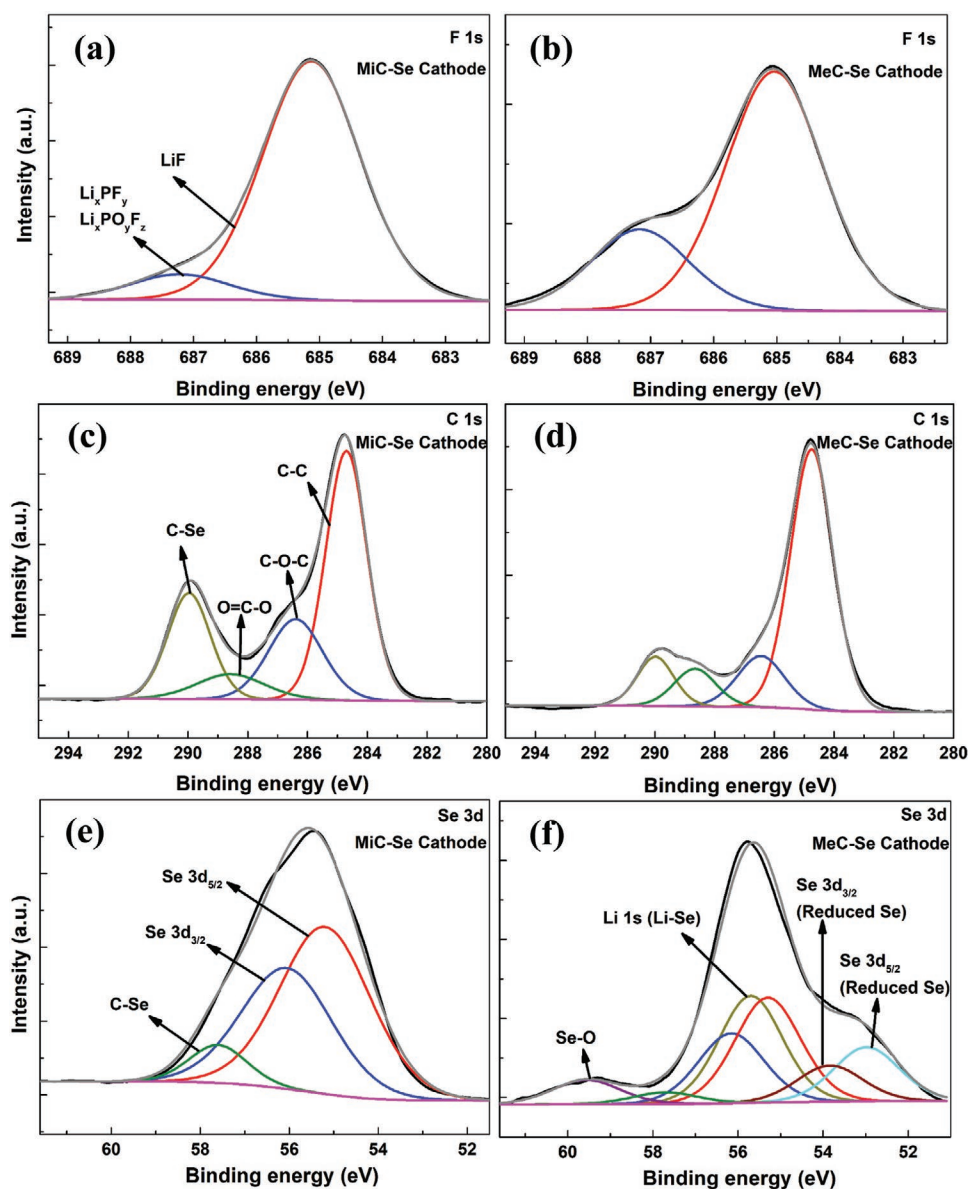


Figure 6. Surface chemistry properties of cyclic MiC/Se and MeC/Se cathode after the fifth cycle in carbonate-based electrolyte. a,b) F 1s, c,d) C 1s, and e,f) Se 3d XPS spectra.

Coulombic efficiency of MiC/Se cathode in ether-based electrolyte is very low due to the redox shuttle of soluble lithium polyselenide between Se cathode and Li metal anode (Figure S9, Supporting Information).^[4,7,13] In the solvate ionic liquid electrolyte, the MiC/Se cathode shows similar electrochemical behavior to that in carbonate-based electrolyte, including a single discharge plateau at 1.8 V in the galvanostatic discharge-charge profiles and an initial capacity of 1010 mAh g⁻¹ (Figure 8b). After 100 cycles, the MiC/Se cathode maintains a capacity of 515 mAh g⁻¹, which is close to the value of MiC/Se cathode in carbonate-base electrolyte (Figure 8c). Because of the high LiTFSI concentration, the viscosity of solvate ionic liquid electrolyte is much higher than that of carbonate-based electrolyte, which may result in a relatively low ionic conductivity. However, the electrode kinetics of

MiC/Se cathode are almost not affected by the viscosity of electrolyte.

EIS of cell using ether and solvate ionic-liquid electrolyte were further measured to know the influence of electrolyte in the electrochemical behaviors of MiC/Se cathode. As shown in Figure S11 in the Supporting Information, the pristine MiC/Se cathode shows the similar Nyquist plots in both ether and solvate ionic-liquid electrolyte. However, after cycling, the impedance responses in the high-middle frequency region were divided into two depressed semicircles. After fitting, the former semicircle in both plots can be also attributed to the interfacial resistance (R_1) including the charge transfer resistance and SEI resistance. The value of R_1 (679 Ω) for the ether electrolyte is much large than that of solvated ionic liquid electrolyte (8.7 Ω). For ether electrolyte, the second semicircle

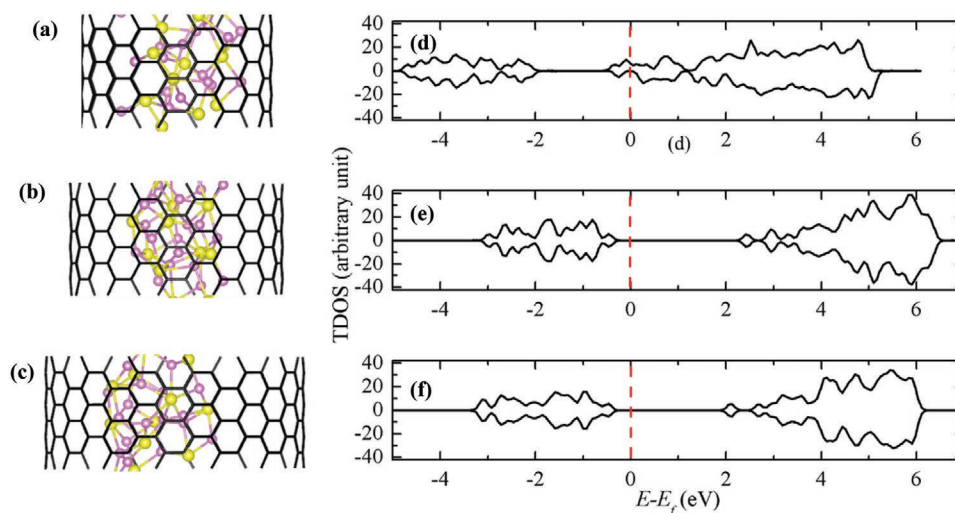


Figure 7. First-principle calculations of electronic structure of $\text{MiC}/\text{Li}_{16}\text{Se}_8$ composite. a–c) Atomic structures of $\text{Li}_{16}\text{Se}_8$ confined within (10, 10) SWCNT, (12, 12) SWCNT, and (14, 14) SWCNT, respectively. d–f) Total density of states of $\text{Li}_{16}\text{Se}_8$ confined within (10, 10) SWCNT, (12, 12) SWCNT, and (14, 14) SWCNT, respectively. Yellow spheres represent Se atoms and violet spheres represent Li atoms.

may be ascribed to the accumulation of electronic/ionic insulating $\text{Li}_2\text{Se}_2/\text{Li}_2\text{Se}$ on cathode surface.^[48,49] The dead $\text{Li}_2\text{Se}_2/\text{Li}_2\text{Se}$ are formed due to the shuttle effect of lithium polyselenide. While for solvated ionic liquid electrolyte, the R_2 may correspond to the desolvation of $[\text{Li}(\text{G}_4)]^+$ cations on the SEI/electrolyte interface.^[50] It is worth noting that the solution resistance (R_s) of ether electrolyte increases from 2.2 to 6.3 Ω , because the dissolution of lithium polyselenides will increase the viscosity of electrolyte. In contrast, the R_s for

solvated ionic liquid electrolyte only shows a slight increase from 9.3 to 10.1 Ω .

To reveal the reason for the fact that MiC/Se cathode shows similar electrochemical behaviors in carbonate and solvate ionic-liquid electrolytes, the theoretical calculations are also performed. **Figure 9** shows the optimized geometries of solvent molecules in various electrolytes. The largest dimensions of ethylene carbonate (EC) and dimethyl carbonate (DMC) are calculated to be 6.25 and 7.96 Å, respectively, which are both

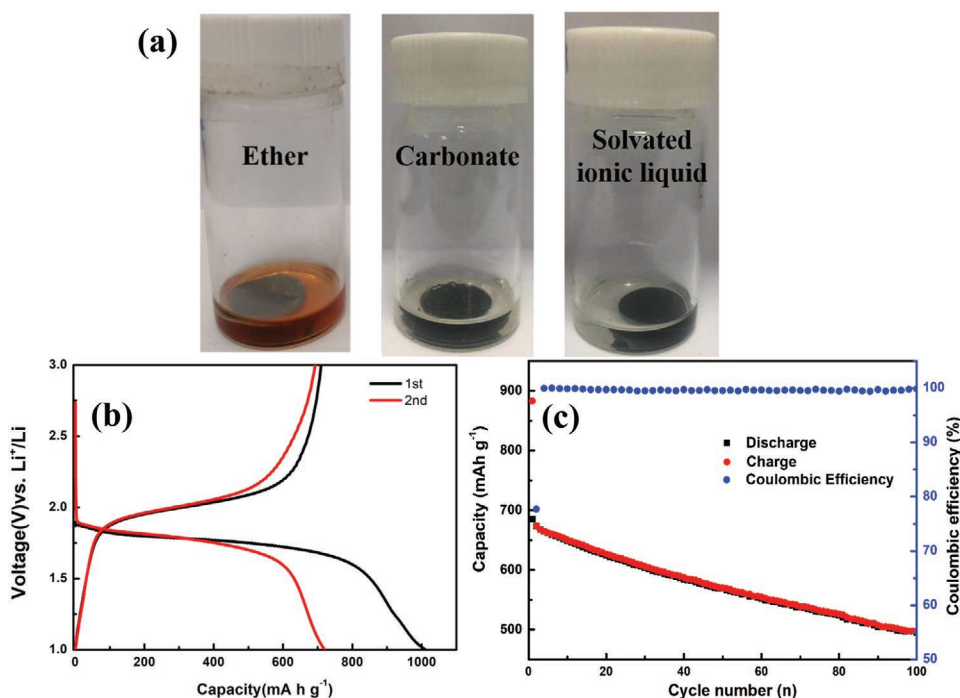


Figure 8. Electrochemical performance of MiC/Se cathode in different electrolytes. a) Digital images of the cycled MiC/Se cathode soaking in the various electrolytes for 12 h. b) Voltage profiles and c) cycling performance and Coulombic efficiency of MiC/Se cathode in solvate ionic liquid electrolyte at 0.5 C.

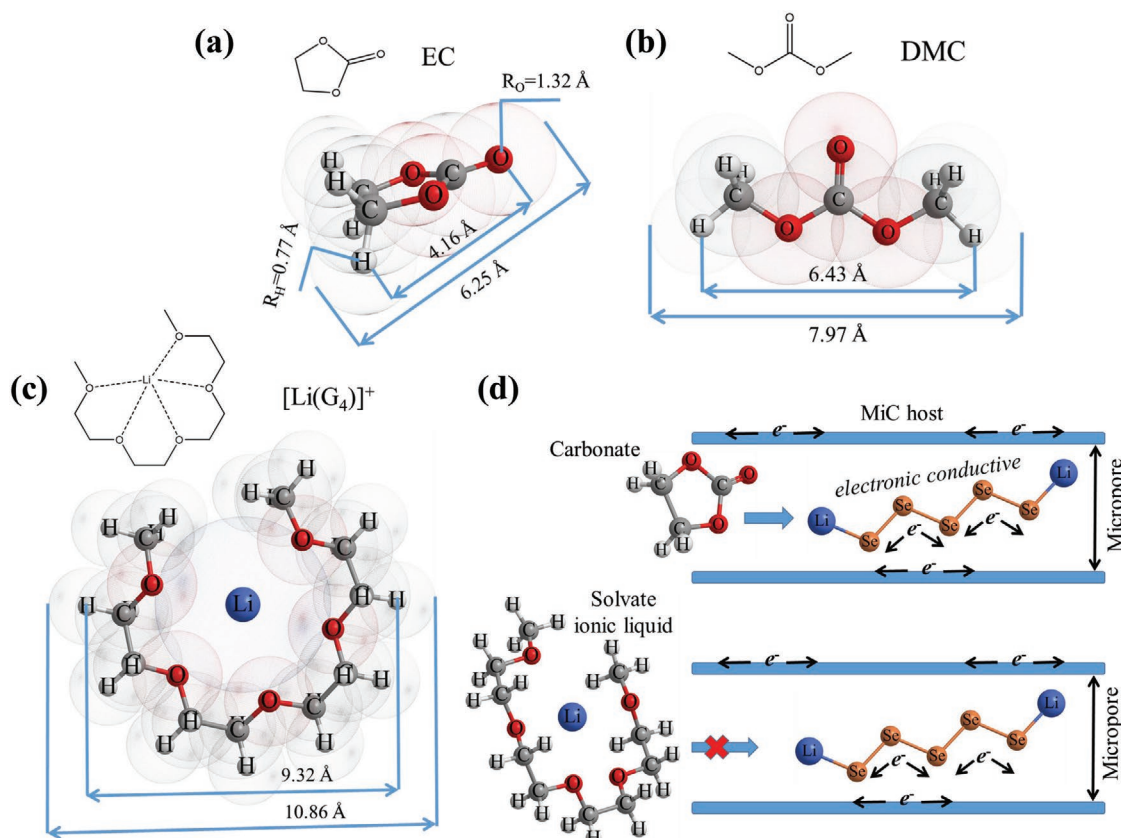


Figure 9. Theoretical calculation of the dimensions of solvent molecules. a) EC, b) DMC, and c) $[\text{Li}(\text{G}_4)]^+$ cation. d) Schematic illustration of lithiation/delithiation process of the MiC/Se cathode in various electrolytes.

smaller than the pore size of MiC materials ($10.20 \text{ \AA}$). The carbonate molecules can easily penetrate into the micropores of MiC materials. In the case of solvate ionic liquid, all the Li^+ ions are coordinated with the tetraglyme (G_4) molecules by Li-O interactions, and the diameter of solvate Li^+ species- $[\text{Li}(\text{G}_4)]^+$ is $10.86 \text{ \AA}$. The micropores of MiC cannot accommodate the $[\text{Li}(\text{G}_4)]^+$ complexes. Thus, as shown in Figure 9d, we speculate that 1) the SEI film is formed on the surface of MiC/Se cathode at the first discharge process; 2) in carbonate-based electrolyte, SEI film that covers the pores is vital to protect the Se chain in micropores and prevent the irreversible reaction between nucleophilic Se anions and the carbonyl group of carbonate solvent;^[4,5] 3) in the charge/discharge process after the first cycle, the Se chains in micropores react with the Li^+ ions which are desolvated at the SEI/electrolyte interface in both carbonate and solvate ionic liquid electrolytes; 4) the electrochemical performance of MiC/Se cathode is mainly determined by the ion conduction and electron transfer in SEI layer and micropores. As exhibited in Figure S12 in the Supporting Information, the ether solvent molecules that are 1,2-dimethoxyethane (DME) and 1,3-dioxolane (DOL) can also enter the interior of micropores of MiC. The low Coulombic efficiency of MiC/Se cathode in ether-based electrolyte indicates that the thin SEI film of 5–6 nm may be not able to block the dissolution of lithium polyselenides (Figure S13, Supporting Information). Future work to explain this phenomenon at molecular level

is undergoing so that more proper electrolyte for the MiC/Se can be developed.

3. Conclusion

In summary, we have systematically investigated the electrochemical properties of Se confined in both MiC and mesoporous carbon (MeC), and we demonstrate that the better electrochemical performance of MiC/Se cathode than MeC/Se cathode can be well explained by its higher ionic and electronic conductivities. By forming a highly stable and Li^+ permeable SEI film on the MiC/Se cathode, the Li^+ transfer in the MiC/Se cathode is accelerated. Theoretical calculations reveal that the spatial confinement of micropores in MiC materials leads to the generation of itinerant electrons, which make MiC to exhibit metallic electronic conductivity. In addition, MiC/Se cathode shows similar electrochemical properties in both carbonate-based and solvate ionic liquid electrolytes with the different size of solvated Li^+ species, which implies the key role that SEI layer plays in the operation of micropore confined selenium cathode by a quasi-solid-solid mechanism. Taken together, this systematic study not only clarifies the governing factor for the electrochemistry of micropores confined selenium, but also provides a new path for the development of high-energy and high-power Li-Se battery.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

X.W., Y.T., and Z.L. contributed equally to this work. The authors are grateful to National Natural Science Foundation of China (Grant Nos. 51772089, 21872046, 51802092, and 51904098), Youth 1000 Talent Program of China, Fundamental Research Funds for the Central Universities, Outstanding Youth Scientist Foundation of Hunan Province (Grant No. 2018JJ1009), Natural Science Foundation of Hunan Province (Grant No. 2019JJ50050) and Provincial Science and Technology Innovation Platform and Talent Plan-Changsha, Zhuzhou and Xiangtan High-level Talents Accumulation Project (Grant No. 2017XK2023), and the Research and Development Plan of Key Areas in Hunan Province (Grant No. 2019GK2235).

Conflict of Interest

The authors declare no conflict of interest.

Keywords

conductivity, confined Se cathodes, electrolytes, lithium–selenium batteries

Received: January 14, 2020

Revised: February 22, 2020

Published online:

- [1] A. Manthiram, Y. Fu, S. H. Chung, C. Zu, Y. S. Su, *Chem. Rev.* **2014**, *114*, 11751.
- [2] C. P. Yang, S. Xin, Y. X. Yin, H. Ye, J. Zhang, Y. G. Guo, *Angew. Chem., Int. Ed.* **2013**, *52*, 8363.
- [3] A. Abouimrane, D. Dambournet, K. W. Chapman, P. J. Chupas, W. Weng, K. Amine, *J. Am. Chem. Soc.* **2012**, *134*, 4505.
- [4] Y. Cui, A. Abouimrane, J. Lu, T. Bolin, Y. Ren, W. Weng, C. Sun, V. A. Maroni, S. M. Heald, K. Amine, *J. Am. Chem. Soc.* **2013**, *135*, 8047.
- [5] C. P. Yang, Y. X. Yin, Y. G. Guo, *J. Phys. Chem. Lett.* **2015**, *6*, 256.
- [6] G. L. Xu, J. Liu, R. Amine, Z. Chen, K. Amine, *ACS Energy Lett.* **2017**, *2*, 605.
- [7] Y. Cui, A. Abouimrane, C. J. Sun, Y. Ren, K. Amine, *Chem. Commun.* **2014**, *50*, 5576.
- [8] L. Liu, Y. Hou, X. Wu, S. Xiao, Z. Chang, Y. Yang, Y. Wu, *Chem. Commun.* **2013**, *49*, 11515.
- [9] D. Kundu, F. Krumeich, R. Nesper, *J. Power Sources* **2013**, *236*, 112.
- [10] Q. Li, H. Liu, Z. Yao, J. Cheng, T. Li, Y. Li, C. Wolverton, J. Wu, V. P. Dravid, *ACS Nano* **2016**, *10*, 8788.
- [11] S. Xin, L. Yu, Y. You, H. P. Cong, Y. X. Yin, X. L. Du, Y. G. Guo, S. H. Yu, Y. Cui, J. B. Goodenough, *Nano Lett.* **2016**, *16*, 4560.
- [12] T. Liu, Y. Zhang, J. Hou, S. Lu, J. Jiang, M. Xu, *RSC Adv.* **2015**, *5*, 84038.
- [13] K. Han, Z. Liu, J. Shen, Y. Lin, F. Dai, H. Ye, *Adv. Funct. Mater.* **2015**, *25*, 455.
- [14] C. Luo, Y. Xu, Y. Zhu, Y. Liu, S. Zheng, Y. Liu, A. Langrock, C. Wang, *ACS Nano* **2013**, *7*, 8003.
- [15] M. Jia, S. Lu, Y. Chen, T. Liu, J. Han, B. Shen, X. Wu, S. J. Bao, J. Jiang, M. Xu, *J. Power Sources* **2017**, *367*, 17.
- [16] T. Liu, C. Dai, M. Jia, D. Liu, S. Bao, J. Jiang, M. Xu, C. M. Li, *ACS Appl. Mater. Interfaces* **2016**, *8*, 16063.
- [17] Z. Li, L. Yin, *Nanoscale* **2015**, *7*, 9597.
- [18] Y. Liu, L. Si, X. Zhou, X. Liu, Y. Xu, J. Bao, Z. Dai, *J. Mater. Chem. A* **2014**, *2*, 17735.
- [19] Z. Li, L. Yin, *ACS Appl. Mater. Interfaces* **2015**, *7*, 4029.
- [20] J. Zhou, J. Yang, Z. Xu, T. Zhang, Z. Chen, J. Wang, *J. Mater. Chem. A* **2017**, *5*, 9350.
- [21] S. F. Zhang, W. P. Wang, S. Xin, H. Ye, Y. X. Yin, Y. G. Guo, *ACS Appl. Mater. Interfaces* **2017**, *9*, 8759.
- [22] Y. J. Hong, Y. C. Kang, *Carbon* **2017**, *111*, 198.
- [23] Z. Li, L. Yuan, Z. Yi, Y. Liu, Y. Huang, *Nano Energy* **2014**, *9*, 229.
- [24] K. Dokko, N. Tachikawa, K. Yamauchi, M. Tsuchiya, A. Yamazaki, E. Takashima, J.-W. Park, K. Ueno, N. Serizawa, S. Seki, M. Watanabe, *J. Electrochem. Soc.* **2013**, *160*, A1304.
- [25] H. Lu, Y. Yuan, Z. Hou, Y. Lai, K. Zhang, Y. Liu, *RSC Adv.* **2016**, *6*, 18186.
- [26] K. Ueno, J. W. Park, A. Yamazaki, T. Mandai, N. Tachikawa, K. Dokko, M. Watanabe, *J. Phys. Chem. C* **2013**, *117*, 20509.
- [27] C. Zheng, M. Liu, W. Chen, L. Zeng, M. Wei, *J. Mater. Chem. A* **2016**, *4*, 13646.
- [28] X. Ji, K. T. Lee, L. F. Nazar, *Nat. Mater.* **2009**, *8*, 500.
- [29] I. L. Li, S. C. Ruan, Z. M. Li, J. P. Zhai, Z. K. Tang, *Appl. Phys. Lett.* **2005**, *87*, 071902.
- [30] J. Zhou, J. Yang, Z. Xu, T. Zhang, Z. Chen, J. Wang, *J. Mater. Chem. A* **2017**, *5*, 9350.
- [31] Y. Yao, M. Chen, R. Xu, S. Zeng, H. Yang, S. Ye, F. Liu, X. Wu, Y. Yu, *Adv. Mater.* **2018**, *30*, 1805324.
- [32] S. K. Park, J. S. Park, Y. C. Kang, *ACS Appl. Mater. Interfaces* **2018**, *10*, 16531.
- [33] J. T. Lee, H. Kim, M. Oschatz, D. C. Lee, F. Wu, H. T. Lin, B. Zdyrko, W. Il Cho, S. Kaskel, G. Yushin, *Adv. Energy Mater.* **2015**, *5*, 1400981.
- [34] M. H. Aboonassr Shiraz, H. Zhu, Y. Liu, X. Sun, J. Liu, *J. Power Sources* **2019**, *438*, 227059.
- [35] X. Li, X. Li, J. Liang, C. Wang, J. Luo, R. Li, X. Sun, *Energy Environ. Sci.* **2018**, *11*, 2828.
- [36] X. Yang, X. Li, K. Adair, H. Zhang, X. Sun, *Structural Design of Lithium–Sulfur Batteries: From Fundamental Research to Practical Application*, Springer, Singapore **2018**.
- [37] X. Li, J. Liang, J. Luo, C. Wang, X. Li, Q. Sun, R. Li, L. Zhang, R. Yang, S. Lu, H. Huang, X. Sun, *Adv. Mater.* **2019**, *31*, 1808100.
- [38] X. Wang, Z. Zhang, Y. Qu, G. Wang, Y. Lai, J. Li, *J. Power Sources* **2015**, *287*, 247.
- [39] J. Wu, N. You, X. Li, H. Zeng, S. Li, Z. Xue, Y. Ye, X. Xie, *J. Mater. Chem. A* **2019**, *7*, 7644.
- [40] J. Q. Huang, T. Z. Zhuang, Q. Zhang, H. J. Peng, C. M. Chen, F. Wei, *ACS Nano* **2015**, *9*, 3002.
- [41] X. Tao, J. Wan, C. Liu, H. Wang, H. Yao, G. Zheng, Z. W. Seh, Q. Cai, W. Li, G. Zhou, C. Zu, Y. Cui, *Nat. Commun.* **2016**, *7*, 11203.
- [42] P. X. Hou, H. Orikasa, T. Yamazaki, K. Matsuoka, A. Tomita, N. Setoyama, Y. Fukushima, T. Kyotani, *Chem. Mater.* **2005**, *17*, 5187.
- [43] E. Markevich, G. Salitra, Y. Talyosef, F. Chesneau, D. Aurbach, *J. Electrochem. Soc.* **2017**, *164*, A6244.
- [44] H. Shin, J. Park, S. Han, A. M. Sastry, W. Lu, *J. Power Sources* **2015**, *277*, 169.
- [45] L. Sha, P. Gao, X. Ren, Q. Chi, Y. Chen, P. Yang, *Chem. - Eur. J.* **2018**, *24*, 2151.
- [46] X. Zhao, L. Yin, T. Zhang, M. Zhang, Z. Fang, C. Wang, Y. Wei, G. Chen, D. Zhang, Z. Sun, F. Li, *Nano Energy* **2018**, *49*, 137.
- [47] L. Wang, X. Zhang, L. Deng, J. Tang, H. Deng, W. Hu, Z. Liu, *ACS Appl. Mater. Interfaces* **2019**, *11*, 4995.
- [48] L. Yuan, X. Qiu, L. Chen, W. Zhu, *J. Power Sources* **2009**, *189*, 127.
- [49] N. A. Cañas, K. Hirose, B. Pascucci, N. Wagner, K. A. Friedrich, R. Hiesgen, *Electrochim. Acta* **2013**, *97*, 42.
- [50] C. Fu, M. B. Oviedo, Y. Zhu, A. Von Wald Cresce, K. Xu, G. Li, M. E. Itkis, R. C. Haddon, M. Chi, Y. Han, B. M. Wong, J. Guo, *ACS Nano* **2018**, *12*, 9775.